

Data Interpretation Exercise
Data Interpretation Assessment (R Programming)

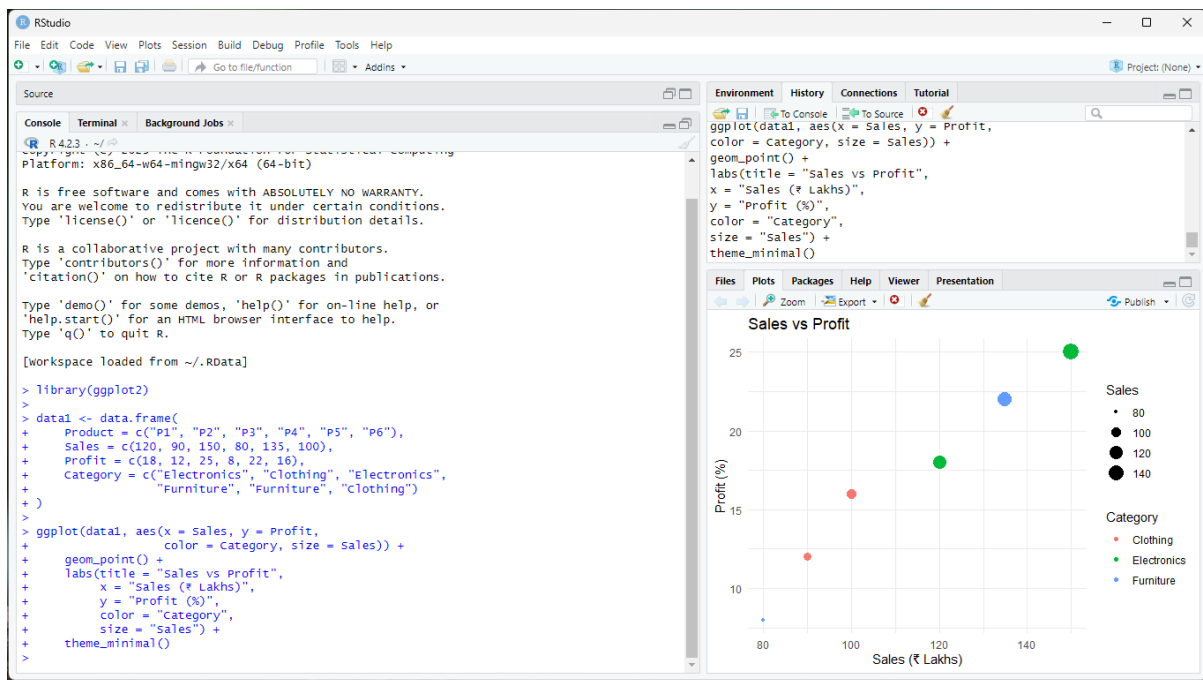
Q1

Code

```
library(ggplot2)

data <- data.frame(
  Product = c("P1", "P2", "P3", "P4", "P5", "P6"),
  Sales = c(120, 90, 150, 80, 135, 100),
  Profit = c(18, 12, 25, 8, 22, 16),
  Category = c("Electronics", "Clothing", "Electronics",
               "Furniture", "Furniture", "Clothing")
)

ggplot(data, aes(x=Sales, y=Profit,
                 color=Category, size=Sales)) +
  geom_point() +
  labs(title="Sales vs Profit",
       x="Sales (₹ Lakhs)",
       y="Profit (%)") +
  theme_minimal()
```



Q2

Code

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Direction = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"),
```

```
  Frequency = c(18, 12, 20, 15, 22, 11, 16, 14)
```

```
)
```

```
ggplot(data, aes(x=Direction, y=Frequency)) +
```

```
  geom_col(fill="steelblue") +
```

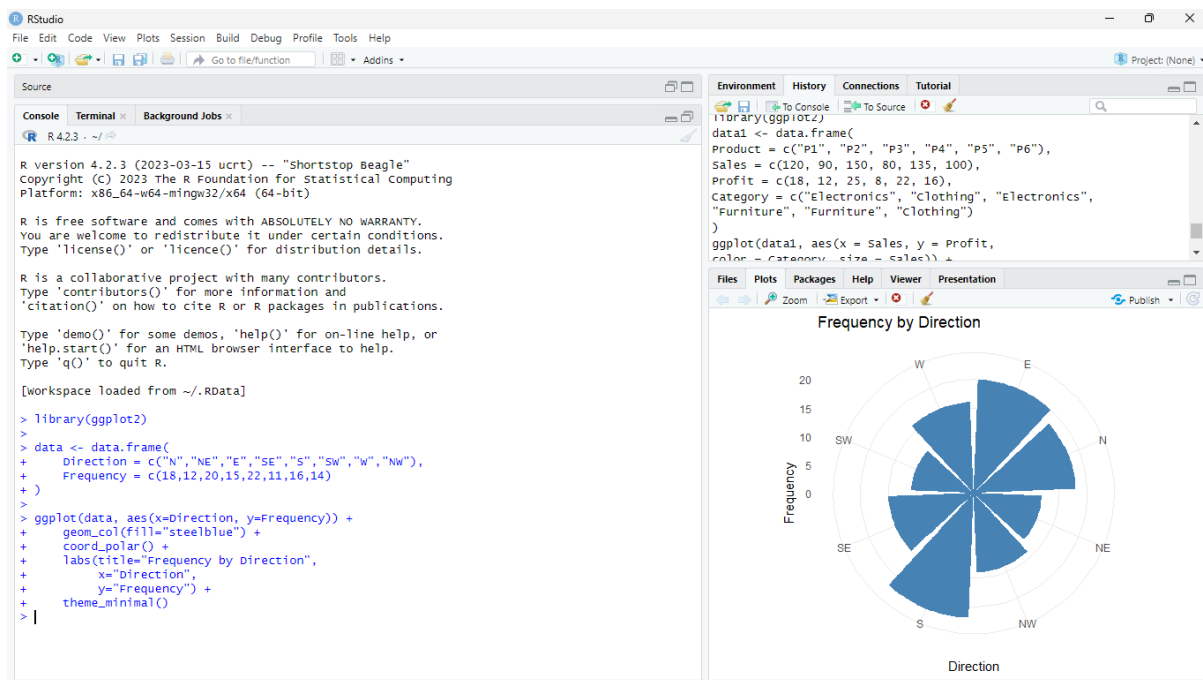
```
  coord_polar() +
```

```
  labs(title="Frequency by Direction",
```

```
        x="Direction",
```

```
        y="Frequency") +
```

```
  theme_minimal())
```



Q3

Code

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Student = c("A","B","C","D","E","F"),
```

```
  Marks = c(45,58,72,81,67,90)
```

```
)
```

```
ggplot(data, aes(x=Student, y=Marks, fill=Marks)) +
```

```
  geom_col() +
```

```
  scale_fill_viridis_c() +
```

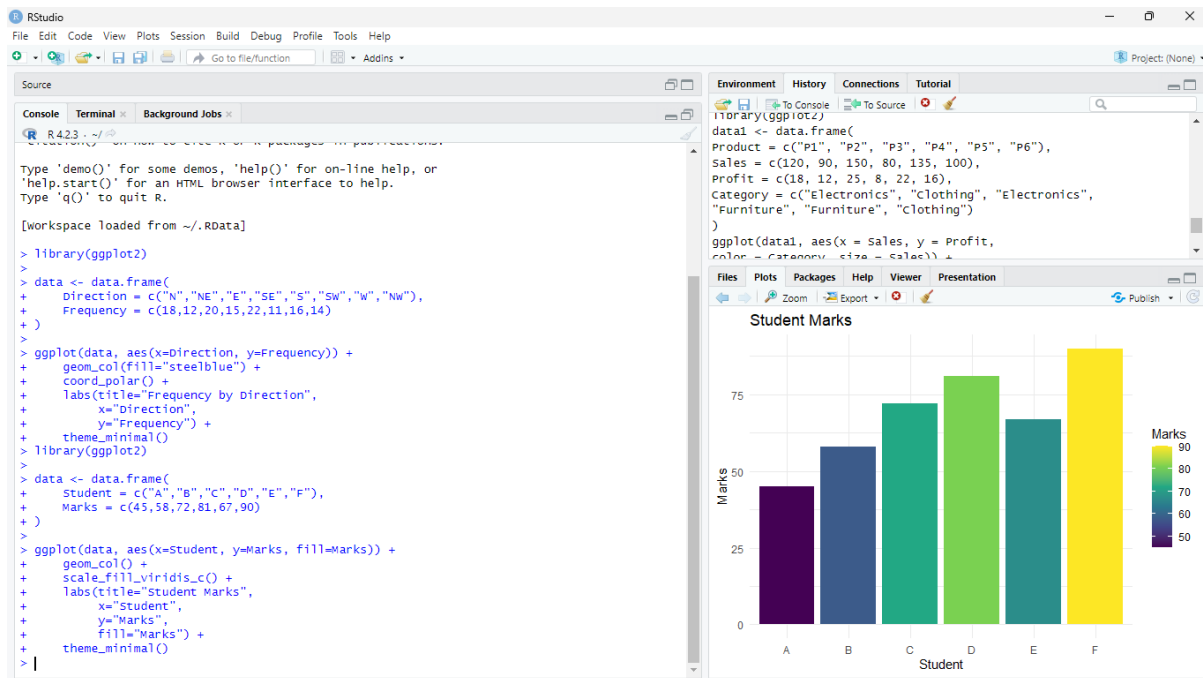
```
  labs(title="Student Marks",
```

```
        x="Student",
```

```
        y="Marks",
```

```
        fill="Marks") +
```

```
  theme_minimal()
```



Q4

Code

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Region = c("North", "South", "East", "West", "Central"),
```

```
  Revenue = c(820, 760, 790, 1010, 770)
```

```
)
```

```
data$Highlight <- ifelse(data$Revenue == max(data$Revenue),
```

```
  "Highest", "Other")
```

```
ggplot(data, aes(x=Region, y=Revenue, fill=Highlight)) +
```

```
  geom_col() +
```

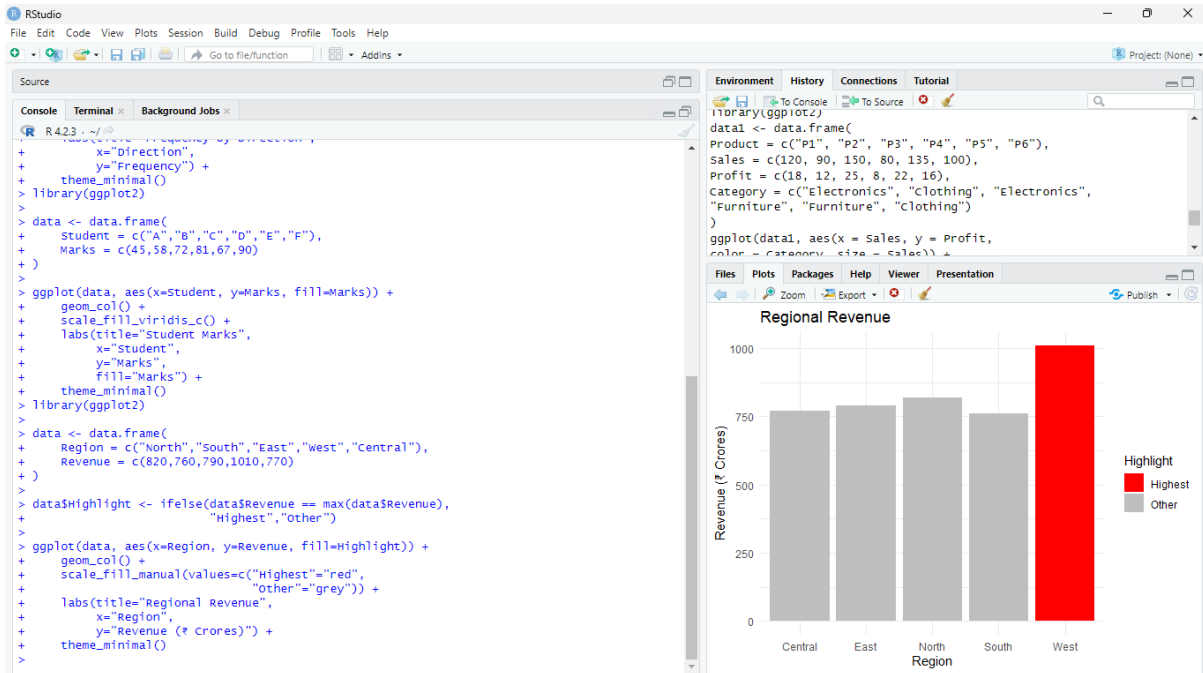
```
  scale_fill_manual(values=c("Highest"="red",
```

```
    "Other"="grey")) +
```

```
  labs(title="Regional Revenue",
```

```
    x="Region",
```

```
theme_minimal()
```



Q5

Code

```
library(ggplot2)
```

```
data <- data.frame(  
  City = c("A", "B", "C", "D", "E"),  
  Population = c(8.2, 5.6, 9.4, 6.8, 4.9),  
  AQI = c(110, 180, 90, 240, 150),  
  Rainfall = c(850, 430, 1200, 520, 680),  
  Region = c("North", "North", "South", "East", "West")  
)
```

```
ggplot(data, aes(x=Population,  
y=AQI,  
size=Rainfall,  
color=Region)) +
```

```

geom_point(alpha=0.8) +
geom_text(aes(label=City), vjust=-1) +
labs(title="Population, AQI and Rainfall",
      x="Population (Million)",
      y="AQI",
      size="Rainfall (mm)",
      color="Region") +
theme_minimal()

```

